1 SURVEY AND EXPERIMENTAL INSTRUMENTS FOR MEASURING RSI PARAMETERS

This appendix suggests simple empirical instruments, survey items and experimental designs, that AI frontier labs could implement within their teams to directly inform the empirical objects in Section **??**. The goal is to measure the return to effective R&D effort, the substitutability between human labor and experimental compute, the extent to which AI capabilities raise effective R&D effort, and whether new bottlenecks emerge as AI systems take over larger parts of the AI R&D process.

1.1 Marginal product of experimental compute

A key object to measure is the marginal product of experimental compute and whether experimental compute is a binding input to effective R&D effort. This can be measured by the following item. In an incentivized variant,fFor a randomly selected respondent or team, implement the corresponding compute increment for one quarter. This turns the answer into a BDM-style elicitation of the team's willingness to use additional experimental compute.

Survey/Experimental item. "Suppose your team's experimental compute budget were doubled next quarter, with no other input changes. By what percentage would your team's research output increase?"

The answer to this question could come in a cardinal or ordinal format.

1.2 Labor–compute substitutability

Survey item. Holding research-output quality fixed, what fraction of your team's monthly working hours would you be willing to trade for an increase in your team's experimental compute budget?

- 0%
- 10%
- 30%
- 50%
- 80%
- I would not trade working hours for additional compute.

Manager scenario item. How much more progress would your team make toward its main research goal under each of the following scenarios?

1. Twice as many researchers, with compute fixed.
2. Twice as much experimental compute, with researchers fixed.
3. Twice as many researchers and twice as much experimental compute.

For each scenario, respondents report the expected percentage increase in research output over the next quarter. The comparison between the three answers identifies whether human researchers and experimental compute are substitutes or complements in the relevant region.

1.3 AI-task automation share

Survey item. What fraction of your team's RSI-relevant tasks is currently performed end-to-end by AI agents, with researchers reviewing only at the final sign-off stage?

- 0–10%
- 10–30%
- 30–50%
- 50–80%
- More than 80%

Task categories. Respondents should answer separately for the following task categories:

- experiment design;
- coding and implementation;
- debugging;
- evaluation design;
- data generation or data filtering;
- training-pipeline operation;
- analysis and write-up.

1.4 Verification bottlenecks and task reallocation

Survey item. Over the last six months, how has your team's allocation of hours changed across the following activities?

- writing code;
- verifying AI-generated output;
- debugging AI-generated output;
- designing new features or experiments;
- evaluation design;
- novel research ideation;
- coordination and review.

For each category, report the approximate percentage change in time spent.

Follow-up item. How often is verification of AI-generated work the binding bottleneck to shipping your team's next deliverable?

- Never
- Rarely
- Sometimes
- Often
- Always

1.5 Experimental menu: current and counterfactual

Survey item. List up to three RSI-relevant experiments your team has run this quarter. For each experiment, report:

- the research objective;
- the experimental-compute cost;
- the human labor required;
- whether AI agents were used;
- the main result or discovery.

Counterfactual item. Now suppose your team's experimental compute budget doubled for one quarter, with all other inputs fixed. List up to three experiments you would prioritize. For each experiment, report:

- the expected experimental-compute cost;
- the expected human labor required;
- the expected marginal return;
- the discovery or capability improvement the experiment might unlock.

1.6 Taste-vs-budget diagnostic

Survey item. Suppose your team had access to additional experimental compute for one quarter, with no other input changes. Could your team identify enough high-marginal-return experiments to spend it productively?

- Yes, many such experiments.
- Mostly yes.
- Partly.
- Mostly no.
- No; compute would outstrip our current stock of good ideas.

1.7 Discovery multiplier

Survey item. When a successful RSI-relevant experiment leads to a discovery, how many further experiments does it typically make possible that your team would not otherwise have known to run?

- 0
- 1
- 2–3
- 4–5
- More than 5

1.8 Inference compute for AI R&D

Survey item. What share of your team's R&D compute budget is spent on inference for research assistance, agents, coding, debugging, evaluation, experiment planning, or analysis?

- 0–5%
- 5–10%
- 10–20%
- 20–40%
- 40–60%
- More than 60%

Follow-up item. How has this share changed over the last six months?

- Fallen substantially
- Fallen slightly
- Stayed roughly constant
- Increased slightly
- Increased substantially

1.9 AI researcher productivity uplift

Survey item. Relative to a no-AI counterfactual, by what percentage has AI use increased your team's effective R&D output over the last six months?

- 0–10%
- 10–30%
- 30–50%
- 50–100%
- 100–300%
- More than 300%

Follow-up item. Which channel accounts for most of this productivity effect?

- faster coding;
- faster debugging;
- better experiment design;
- faster evaluation;
- better data generation or filtering;
- faster analysis and write-up;
- autonomous completion of full research tasks;
- other.